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Summary

Software Engineer with hands-on experience building and troubleshooting software applications, working with SQL and NoSQL databases, REST APIs, and automated testing. Strong analytical and problem-solving skills with a methodical approach to debugging, investigating application behavior, validating data, and resolving technical issues. Comfortable learning new tools and technologies and communicating technical concepts clearly.

Skills

- **Programming:** JavaScript (ES6+), TypeScript, C++
- **Databases:** SQL, PostgreSQL, MongoDB, TypeORM, Mongoose
- **Software Development:** Node.js, NestJS, Express.js, React, mongoose, TypeORM
- **Testing & Tools:** Jest, Supertest, Git, Docker, Postman

Education

B.Sc. in Computer Engineering-Faculty of Engineering, Modern Academy

Sept 2018 – May 2023

Projects

Project Management Workspace (Node.js, TypeScript, NestJS, PostgreSQL, TypeORM, Socket.io)

Real-time multi-tenant project management platform supporting secure collaboration and workspace isolation.

- Built APIs using NestJS and PostgreSQL with complex relational models for workspaces, boards, tasks, and users.
- Implemented JWT authentication, RBAC, and custom Guards for secure multi-tenant access control
- Developed real-time task synchronization using Socket.io and applied event-driven architecture for audit logging and modular system design
- GitHub Repository: <https://github.com/Omarmadkour71/project-management-system>

E-commerce Platform – Full RESTful Backend API (Node.js, Express, MongoDB)

A production-style backend for an online store with authentication, cart management and payment simulation.

- Built REST APIs for products, categories, carts, and order processing with validation and business logic
- Integrated Stripe Checkout and webhook workflows with JWT authentication and role-based authorization
- Applied security practices including Helmet, rate limiting, and automated testing using Jest and Supertest
- GitHub Repository: <https://github.com/Omarmadkour71/ecommerce-platform>

Natours – Tours Booking RESTful API (Node.js, Express, MongoDB)

Backend booking platform with authentication, payments, and advanced API functionality.

- Implemented JWT authentication, role-based authorization, password reset workflows, and protected routes
- Built API features including filtering, pagination, geospatial queries, aggregation pipelines, and image processing
- Integrated Stripe payments and applied backend security practices including Helmet, sanitization, and rate limiting
- GitHub Repository: <https://github.com/Omarmadkour71/Natours>

Languages

Arabic (Native), English (Very Good)